

DSA MySQL CAPSTONE PROJECT: OLIST BRAZILLIAN E-COMMERCE DATASET

1. Data Overview

The Olist dataset contains information about orders placed through a Brazilian e-commerce marketplace. It consists of eight related tables covering customers, orders, order items, payments, products, sellers, reviews, and geographic information. The order data covers purchases from September 4, 2016 to October 17, 2018, although September and October 2018 contain limited activity and may not represent complete months. Some products have no recorded category, several delivery dates are missing, many review comments are blank, and the geolocation table contains duplicate records.

2. Business Insights

Monthly revenue increased significantly over the period, with a noticeable peak near the end of 2017. The monthly revenue analysis showed that November 2017 generated the highest revenue at 1,194,882.80. This was an increase of 415,204.92 from October, followed by a decrease of 316,481.32 in December. The results suggest that customer demand may be stronger during major year-end shopping periods, so the business should consider seasonality when planning inventory and promotions.

Revenue comes from several strong product categories rather than one dominant category. Beauty and Health generated the highest product revenue at 1,258,681.34, followed by Watches and Gifts at 1,205,005.68 and Bed, Bath and Table at 1,036,988.68. However, Beauty and Health accounted for only 9.26% of total product revenue. This indicates that Olist has a relatively diversified product mix, although its leading categories still make an important contribution to overall sales.

Most customers purchased only once. The customer-order analysis identified 93,099 one-time buyers and only 2,997 repeat buyers, meaning that repeat buyers represented approximately 3.12% of all customers. This shows that the business relies heavily on attracting new customers instead of generating repeat purchases. Improving customer retention could provide an additional source of revenue while reducing the need to depend entirely on continuous customer acquisition.

3. Recommendations

Prepare inventory and promotional campaigns around the strongest categories and periods. Olist should increase inventory availability and marketing activity for Beauty and Health, Watches and Gifts, and other leading categories before high-demand periods such as November. Product rankings within each category can help determine which specific items should receive priority.

Develop a targeted customer-retention program. Olist could use the Low, Medium, and High spending tiers to send more relevant offers, product recommendations, or repeat-purchase discounts. Medium- and high-spending one-time buyers should be prioritized because they have already demonstrated stronger purchasing potential but have not yet returned for another order.